

ASSIGNMENT REPORT-4

Submitted by :

KARTHIKEYAN M [26120004]

MIRTHULA R [261200121]

Under the guidance of :

HEMA ARUNACHALAM MURTHY

For The Subject

Pattern Recognition and Machine Learning

(26AI5707)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SHIV NADAR UNIVERSITY

KALAVAKAM

SEPTEMBER 2026

INTRODUCTION

[WRITE HERE: In 4–5 lines, explain that the assignment studies how isotropic, diagonal, and full covariance matrices change the shape and orientation of a two-dimensional Gaussian. Also mention that the final experiment compares decision boundaries between two Gaussian classes.]

MATHEMATICAL BACKGROUND

For a d -dimensional Gaussian random vector, the probability density is

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right). \quad (1)$$

The quadratic term

$$d_M^2(\mathbf{x}, \boldsymbol{\mu}) = (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \quad (2)$$

is the squared Mahalanobis distance. The constant density curves are obtained by fixing this value:

$$(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c. \quad (3)$$

If λ_i and $\mathbf{q}_i$ are the eigenvalues and eigenvectors of $\boldsymbol{\Sigma}$, then $\mathbf{q}_i$ gives a principal direction and $\sqrt{\lambda_i}$ gives the standard deviation along that direction. At contour level c , the full axis length in that direction is $2\sqrt{c\lambda_i}$.

[WRITE HERE: Briefly explain why Mahalanobis distance is more suitable than raw Euclidean distance when the two features have different spreads or are correlated.]

EXPERIMENTAL SETUP

The experiments used $N = 1000$ samples and a fixed random seed of 42. Two one-dimensional Gaussian datasets were generated and then stacked as columns to form the two dimensional samples. The initial parameters were

$$\boldsymbol{\mu} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad \sigma = 1. \quad (4)$$

[WRITE HERE: Mention that the Gaussian samples were produced using the supplied Gaussian generator. Explain in one sentence why a fixed seed was used.]

ISOTROPIC COVARIANCE MATRIX

For the isotropic case, the target covariance matrix was

$$\mathbf{C} = \sigma^2 \mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (5)$$

The estimated mean and covariance were

$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} 1.9841 \\ 3.9676 \end{bmatrix}, \quad \hat{\mathbf{C}} = \begin{bmatrix} 1.0380 & 0.0463 \\ 0.0463 & 1.0752 \end{bmatrix}. \quad (6)$$

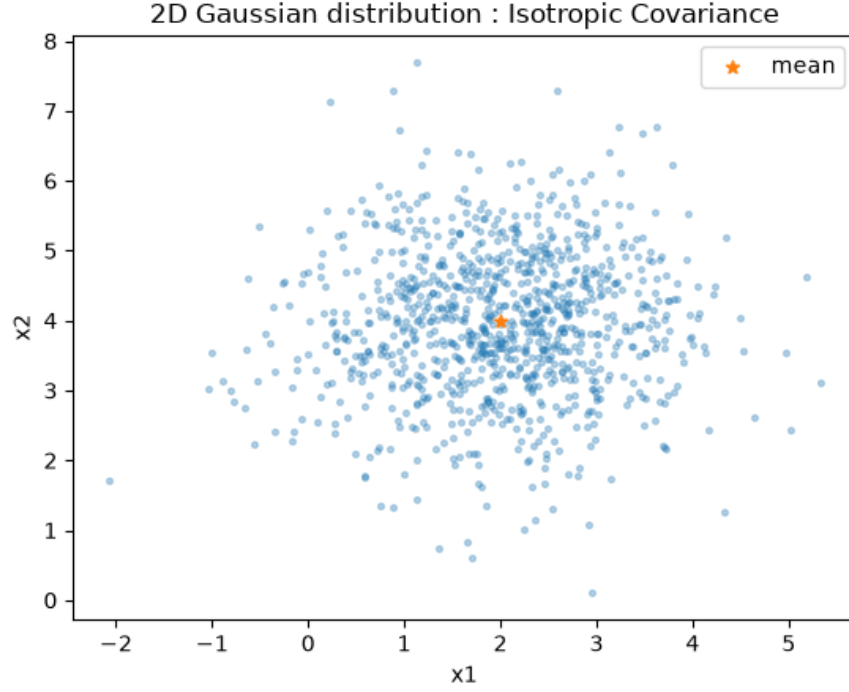


Figure 1: Samples generated using isotropic covariance.

Table 1: Theoretical and estimated eigenvalues for isotropic covariance.

Direction	Theoretical λ	Estimated λ	Estimated axis length ($c = 1$)
Major	1.0000	1.1065	2.104
Minor	1.0000	1.0067	2.007

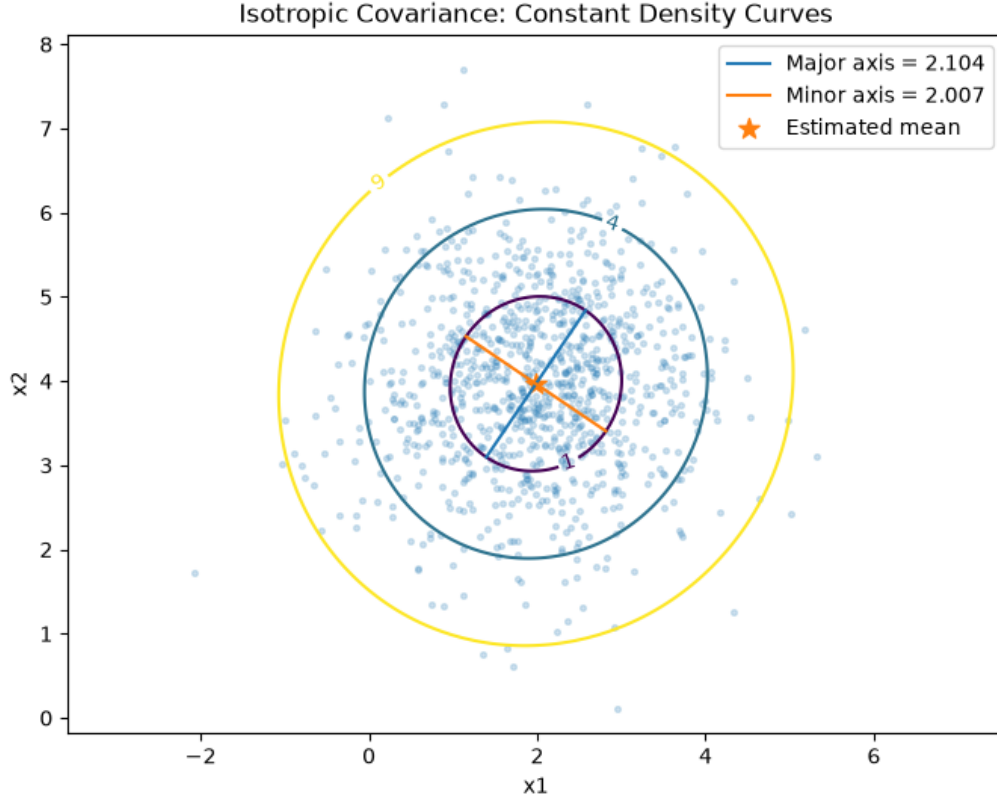


Figure 2: Constant Mahalanobis-distance curves and estimated principal axes for the isotropic data.

[WRITE HERE: Compare the target and estimated covariance/eigenvalues. Explain why the off-diagonal covariance is small but not exactly zero, and why the sampled contour is close to a circle rather than a perfect circle.]

DIAGONAL COVARIANCE MATRIX

The target diagonal covariance was

$$\mathbf{C}_{\text{diag}} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}. \quad (7)$$

Starting from the isotropic data, the transformation used was

$$\mathbf{Y} = \boldsymbol{\mu} + \mathbf{C}_{\text{diag}}^{1/2}(\mathbf{X} - \boldsymbol{\mu}), \quad \mathbf{C}_{\text{diag}}^{1/2} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}. \quad (8)$$

The estimated covariance was

$$\hat{\mathbf{C}}_{\text{diag}} = \begin{bmatrix} 4.1520 & 0.0927 \\ 0.0927 & 1.0752 \end{bmatrix}. \quad (9)$$

Table 2: Theoretical and estimated eigenvalues for diagonal covariance.

Direction	Theoretical λ	Estimated λ	Estimated axis length ($c = 1$)
Major	4.0000	4.1548	4.077
Minor	1.0000	1.0724	2.071

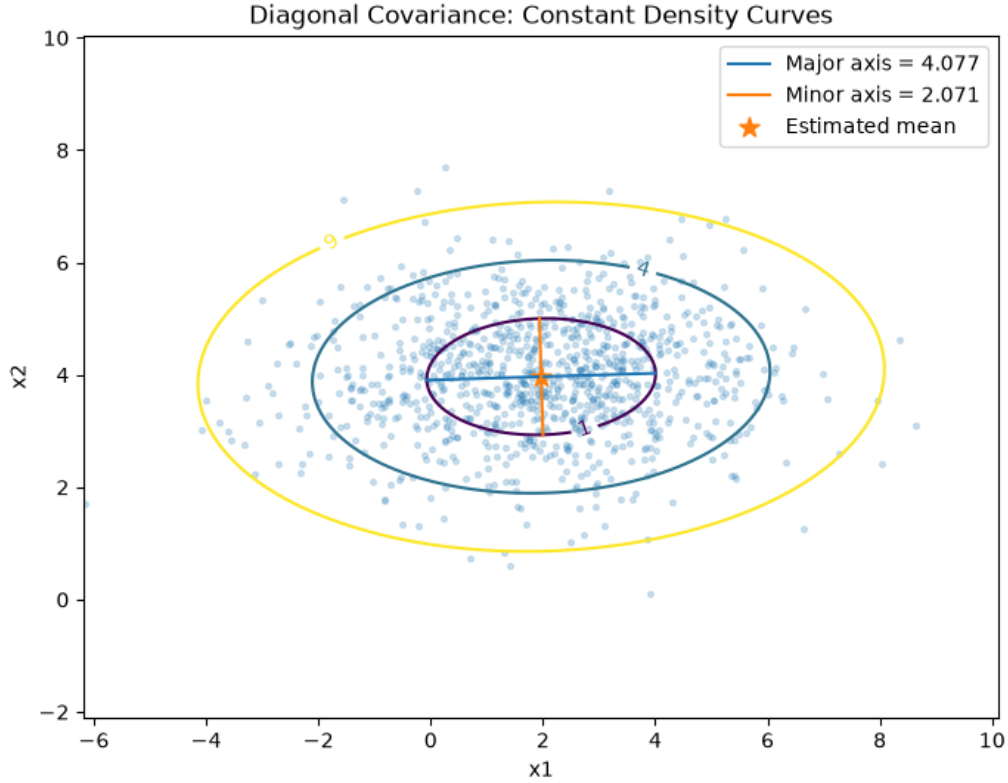


Figure 3: Constant density curves and principal axes for diagonal covariance.

[WRITE HERE: Explain why the first coordinate becomes approximately twice as wide in standard-deviation units, why the contours are elliptical, and why their principal axes remain close to the coordinate axes.]

FULL COVARIANCE MATRIX

The full covariance matrix was chosen as

$$\mathbf{C}_{\text{full}} = \begin{bmatrix} 4.0 & 1.5 \\ 1.5 & 1.0 \end{bmatrix}. \quad (10)$$

For a full covariance matrix, the element-wise square root is not the required matrix square root. Using the eigendecomposition $\mathbf{C} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$, it was constructed as

$$\mathbf{C}^{1/2} = \mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^T, \quad \mathbf{C}^{1/2}(\mathbf{C}^{1/2})^T = \mathbf{C}. \quad (11)$$

The transformed samples had estimated covariance

$$\hat{\mathbf{C}}_{\text{full}} = \begin{bmatrix} 4.2597 & 1.6625 \\ 1.6625 & 1.1065 \end{bmatrix}. \quad (12)$$

Table 3: Theoretical and estimated eigenvalues for full covariance.

Direction	Theoretical λ	Estimated λ	Estimated axis length ($c = 1$)
Major	4.6213	4.9744	4.461
Minor	0.3787	0.3919	1.252

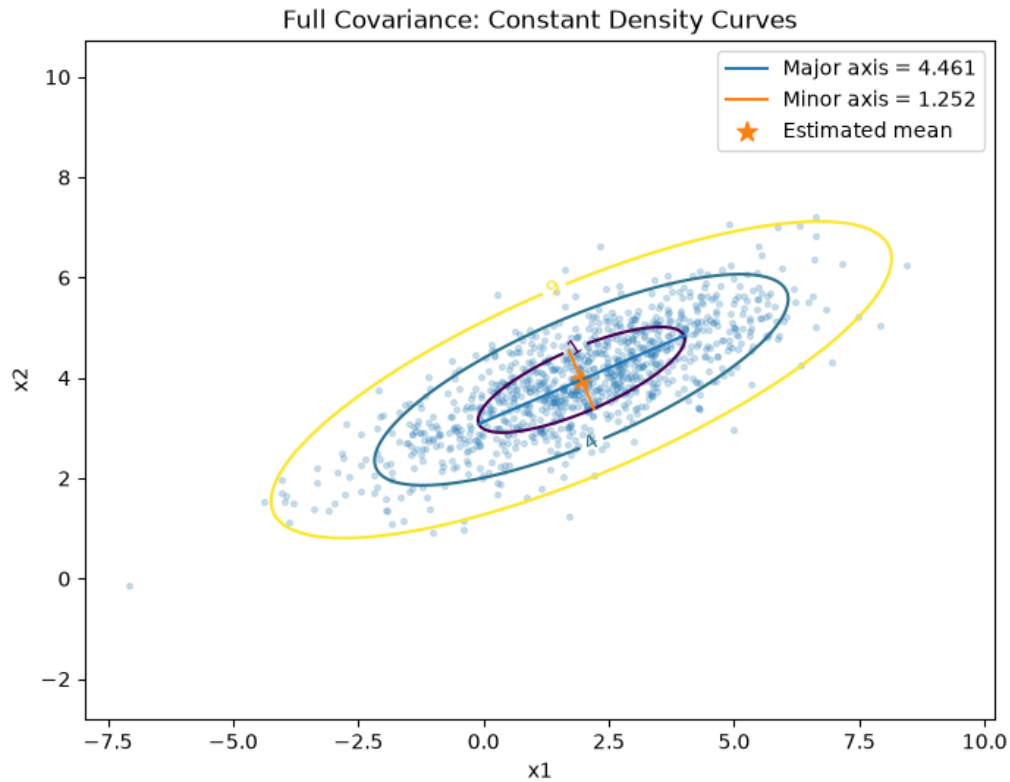


Figure 4: Tilted constant density curves and principal axes for full covariance.

[WRITE HERE: Explain how the nonzero off-diagonal covariance makes the variables move together and rotates the ellipse. Connect the longer direction to the larger eigenvalue and the shorter direction to the smaller eigenvalue.]

TWO GAUSSIAN CLASSES

For equal class priors, the class-dependent part of the Gaussian discriminant function is

$$g_i(\mathbf{x}) = -\frac{1}{2} \log |\Sigma_i| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^\top \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i). \quad (13)$$

The decision boundary contains all points satisfying

$$g_1(\mathbf{x}) - g_2(\mathbf{x}) = 0. \quad (14)$$

Equal covariance matrices

The first experiment used

$$\boldsymbol{\mu}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{\mu}_2 = \begin{bmatrix} 4 \\ 4 \end{bmatrix}, \quad \Sigma_1 = \Sigma_2 = \mathbf{I}. \quad (15)$$

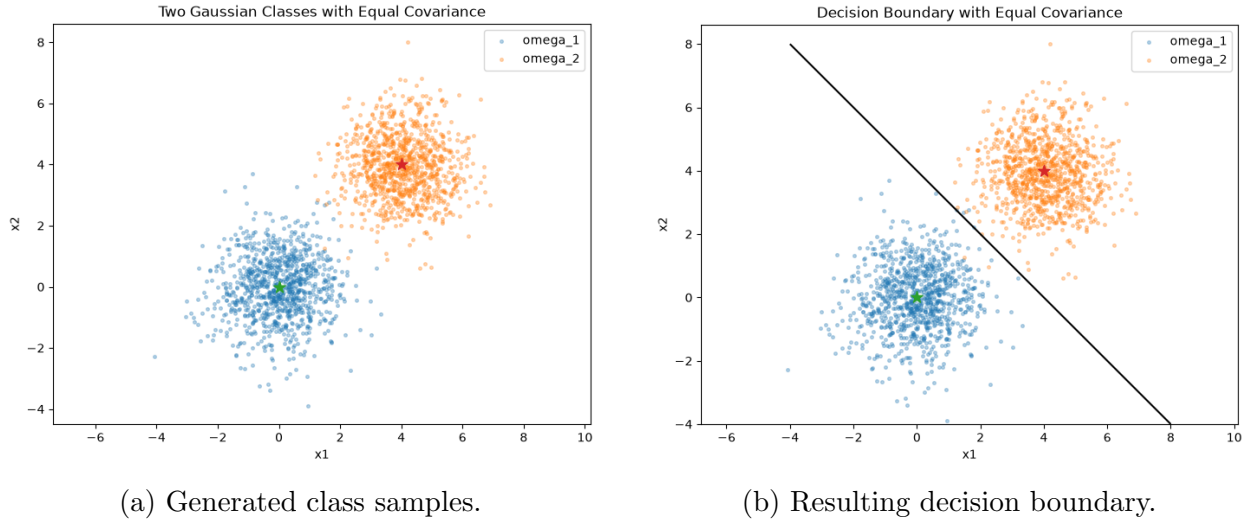


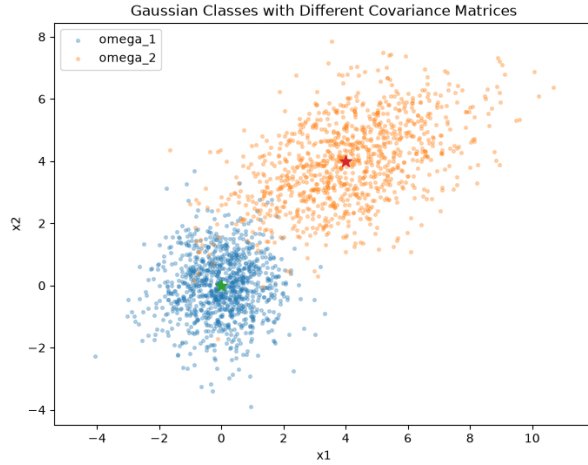
Figure 5: Two Gaussian classes with equal covariance matrices.

[WRITE HERE: Explain why the determinant and quadratic terms common to both classes cancel, leaving a straight-line boundary. For these means, also state why the line passes through the midpoint (2,2).]

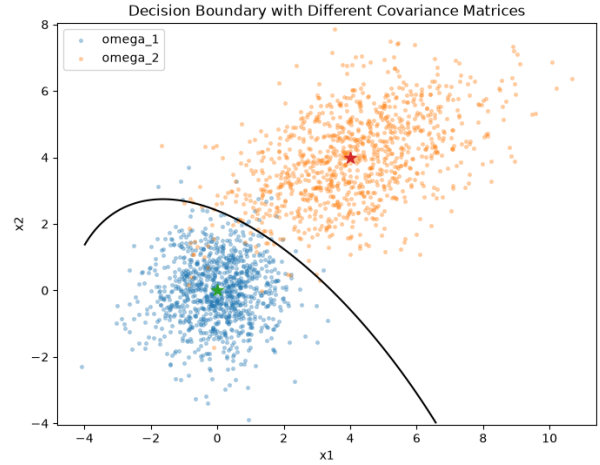
Different covariance matrices

The second experiment retained the same means but used

$$\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \Sigma_2 = \begin{bmatrix} 4 & 1.5 \\ 1.5 & 2 \end{bmatrix}. \quad (16)$$



(a) Generated class samples.



(b) Resulting decision boundary.

Figure 6: Two Gaussian classes with different covariance matrices.

[WRITE HERE: *Explain why the quadratic terms no longer cancel and therefore produce a curved boundary. Relate the curve to the greater spread and tilt of the second class. If two branches appear for another plot range or parameter choice, mention that both are genuine parts of the quadratic boundary.*

INFERENCE

[WRITE HERE: *Summarize the progression: isotropic covariance gives an approximately circular cloud, diagonal covariance changes the axis-wise spreads, full covariance introduces rotation through correlation, equal class covariances give a linear boundary, and unequal covariances give a quadratic boundary.*